

1. Administrative & Study Identification

Full Study Title: A First-in-Human Study to Investigate the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of R07507062 in Participants With Systemic Lupus Erythematosus **Short Title/Acronym:** R07507062 Phase 1 SLE Study **Protocol Number:** BP44315 (EU Trial No: 2022-502632-39-00) **NCT Number:** NCT05835986 **Sponsor Name:** Hoffmann-La Roche (Company: Roche) **Investigational Product:** R07507062 (CD19-targeting T-cell bispecific antibody / CD19TCB) **Phase of Development:** Phase 1 **Medical Monitor:** Head of Regulatory Management / Clinical Operations Lead, Hoffmann-La Roche

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the safety and tolerability of subcutaneously administered R07507062 in participants with systemic lupus erythematosus (SLE), primarily determined by Dose Limiting Adverse Events (DLAEs). **Secondary Objectives:** To evaluate the pharmacokinetics (PK) and pharmacodynamics (PD) of R07507062, including T cell activation, cytokine release, and B cell depletion.

2.2 Background & Rationale Systemic lupus erythematosus (SLE) is an autoimmune disease driven by B cells producing autoantibodies. Despite the success of existing B-cell depletion therapies, efficacy can be limited due to incomplete depletion of tissue-resident B cells. To address this, R07507062 was designed as a next-generation, off-the-shelf T-cell bispecific antibody (CD19TCB) that simultaneously attaches to CD19 on B cells and CD3 on T cells. By activating T cells to eliminate B cells more profoundly, it aims to decrease autoantibody production and limit associated organ damage.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-escalation) **Blinding:** Open-label **Randomization:** N/A (Ascending dose allocation) **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target \$N\$: 70 participants (Estimated) **Number of Sites:** Multicenter study across approximately 13 countries **Estimated Study Start:** 18-Dec-2023 **Estimated Primary Completion:** 30-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to ≤ 70 years. 2. Diagnosis of SLE according to the 2019 EULAR or ACR Classification Criteria at least 24 weeks prior to Screening, and treated for SLE according to standard clinical practice. 3. Presence of anti-double stranded DNA (dsDNA), anti-Smith (Sm), anti-RNP, or anti-SS-A above the upper limit of normal (ULN); or positive ANA ($\geq 1:160$). 4. Active SLE disease, as demonstrated by the Systemic Lupus Erythematosus Disease Activity Index 2000 (SLEDAI-2K) total score of ≥ 4 with at least 1 positive clinical item. 5. Receiving stable doses of standard SLE therapies prior to Screening: - For oral corticosteroids (OCS): ≤ 20 mg/day prednisone or equivalent, stable for at least 7 days prior to Day 1. - For conventional immunosuppressants (e.g., azathioprine, sulfasalazine, mycophenolate mofetil [≤ 3.0 g/day], mycophenolic acid [≤ 3 g/day], methotrexate, calcineurin inhibitors): stable dose for at least 6 weeks prior to Screening and expected to remain stable during the study.

4.2 Exclusion Criteria 1. Active or unstable lupus-associated neuropsychiatric disease, or organ-threatening SLE manifestations (e.g., active myocarditis) considered severe by the Investigator. 2. Catastrophic or severe antiphospholipid syndrome within 12 months prior to or during Screening. 3. Presence of severe lupus-associated renal disease that is likely to require treatment with cyclophosphamide, B-cell-depleting therapies, or other biologic/targeted therapies. 4. Severe active systemic autoimmune disease other than SLE. 5. Active infection of any kind (excluding fungal infection of nail beds); history of serious recurrent or chronic infection (especially respiratory); active/history of recurring or severe tuberculosis or positive IGRA (treated latent TB allowed); active CMV or EBV; HBV infection or positive serology indicative of current/past infection; HIV positive; or active HCV. 6. Moderate or severe chronic obstructive pulmonary disease (COPD). 7. History of progressive multifocal leukoencephalopathy (PML). 8. History of macrophage-activation syndrome and/or hemophagocytic lymphohistiocytosis. 9. History of cancer within 5 years prior to Screening (except adequately treated/cured basal cell carcinoma, non-melanoma skin cancer, and cervical cancer in situ). 10. Intolerance or contraindication to study therapies, including history of severe allergic/anaphylactic reactions to monoclonal antibodies or hypersensitivity to R07507062 components. 11. Prior treatment with anti-CD19 or anti-CD20 therapy (e.g., blinatumomab, obinutuzumab, rituximab, ocrelizumab, ofatumumab) within 6 months. 12. Prior treatment with JAK/BTK/TYK2 inhibitors (e.g., baricitinib, tofacitinib, upadacitinib, filgotinib, ibrutinib, fenebrutinib) within 30 days. 13. Receipt of cyclophosphamide or biologic therapies (e.g., adalimumab, etanercept, golimumab, infliximab, belimumab, ustekinumab, anifrolumab, secukinumab, atacicept) within 4 weeks. 14. Receipt of any investigational therapy (except SARS-CoV-2 vaccines) within 30 days or 5 drug-elimination half-lives. 15.

Immunoglobulin (IgG) level of < 6 g/L. 16. Estimated glomerular filtration rate (eGFR) < 45 mL/min/1.73 m^2 .

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Conducted in two parts. Part 1 is a single ascending dose-finding (SAD) cohort. Part 2 is a dose escalation with fractionated dosing (up to 3 doses administered in the first month). **Route of Administration:** Subcutaneous (SC) injection for R07507062. Tocilizumab is an additional investigational medicinal product (IMP), which may be used at the investigator's discretion to treat Cytokine Release Syndrome (CRS) if required. **Duration of Treatment:** Short-term fractionated doses within the first month, followed by an extended safety and PK/PD follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of conventional SLE therapies including steroids, hydroxychloroquine, mycophenolate mofetil (MMF), azathioprine, and methotrexate. **Prohibited:** Recent cyclophosphamide or biologic therapies (e.g., adalimumab, belimumab, anifrolumab) within 4 weeks; anti-CD19/CD20 within 6 months.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -1	Informed Consent, SLE Diagnosis Confirmation, Medical History, Labs
Baseline / Dosing	Day 1	First SC Dose, Inpatient Monitoring, Intensive PK/PD sampling
Acute Follow-Up	Days 1 to 3+	Hospital inpatient stay with assessments 6 times on dosing day, then daily for at least 3 days for CRS monitoring
End of Evaluation	Day 29	Primary safety evaluation (DLAE assessment completion), Ongoing follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Part 1:** Number of Participants with Dose Limiting Adverse Events (DLAEs). 2. **Part 2 (Fractionated Dose Cohort):** Number of Participants with DLAEs. 3. **Safety:** Number of Participants with Adverse Events (AEs), reported according to NCI CTCAE v5.0, with Cytokine Release Syndrome (CRS) graded per ASTCT criteria.

7.2 Secondary & Exploratory Endpoints 1. **Pharmacokinetics (PK) of R07507062:** - Serum Concentration of R07507062 - Time to Maximum Serum Concentration (T_{max}) of R07507062 - Maximum Serum Concentration (C_{max}) of R07507062 2. **Pharmacodynamics (PD) of R07507062:** Including T-cell activation, cytokine release markers, and quantification of B-cell depletion.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive inpatient monitoring post-dosing for acute reactions, notably Cytokine Release Syndrome (CRS). Graded per NCI CTCAE v5.0 and ASTCT. **Serious Adverse Events (SAEs):** Captured through continuous tracking; protocol dictates immediate intervention with tocilizumab if severe CRS manifests. **Data Monitoring Committee (DMC):** Internal safety review committees evaluate DLAE rates to approve escalation to the next dose level.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all enrolled participants receiving at least one dose of R07507062). **Sample Size Justification:** $N=70$. The sample size is not based on statistical power but on clinical, standard dose-finding Phase 1 designs to adequately identify DLAEs. **Handling of Missing Data:** Missing PK/PD safety data typically handled without imputation (as observed) for Phase 1 toxicity profiles.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring during the inpatient dosing phases. **Audit Trail:** Standard clinical trial eCRF locking, source data verification, and continuous data tracking for dose-escalation reviews.

11. Study Identifiers & Governance

Primary ID: BP44315 **Registry Number:** NCT05835986 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** R07507062 Phase 1 SLE Study **Official Regulatory Title:** An Open-label, Multicenter, Dose Escalation, First-in-Human Study to Investigate the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of Subcutaneously Administered R07507062 in Participants With Systemic Lupus Erythematosus

12. Clinical Framework (Systemic Lupus Erythematosus Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE) **Diagnosis Threshold:** 2019 EULAR/ACR Criteria + Autoantibody confirmation (ANA $\geq 1:160$, anti-dsDNA, anti-Sm, etc.) + SLEDAI-2K score ≥ 4 **Medical Methodology:** B-cell Depletion via T-cell

engagement **Optimization Target:** Maximum tolerated dose /
Recommended Phase 2 Dose determination

13. Study Procedures & Methodology

Management Pathway: Intensive inpatient observation for CRS post-SC injection **Education Protocol (HCP):** Site training on ASTCT CRS grading and proper administration of tocilizumab rescue therapy **Education Protocol (Patient):** Comprehensive informed consent regarding first-in-human risks and CRS symptom awareness **Quality Audit Mechanism:** Phase 1 cohort-by-cohort dose-escalation review meetings

14. Observation & Follow-Up Schedule

Total Duration: Primary DLAE evaluation spans 28/29 days per cohort, followed by long-term follow-up **Onsite Visit Cadence:** Inpatient for at least 3 days post-dosing, transitioning to scheduled outpatient safety follow-ups **Remote Monitoring Frequency:** As mandated by the protocol post-discharge **Checklist Review Interval:** Daily during acute phase; standard intervals during chronic follow-up

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				